

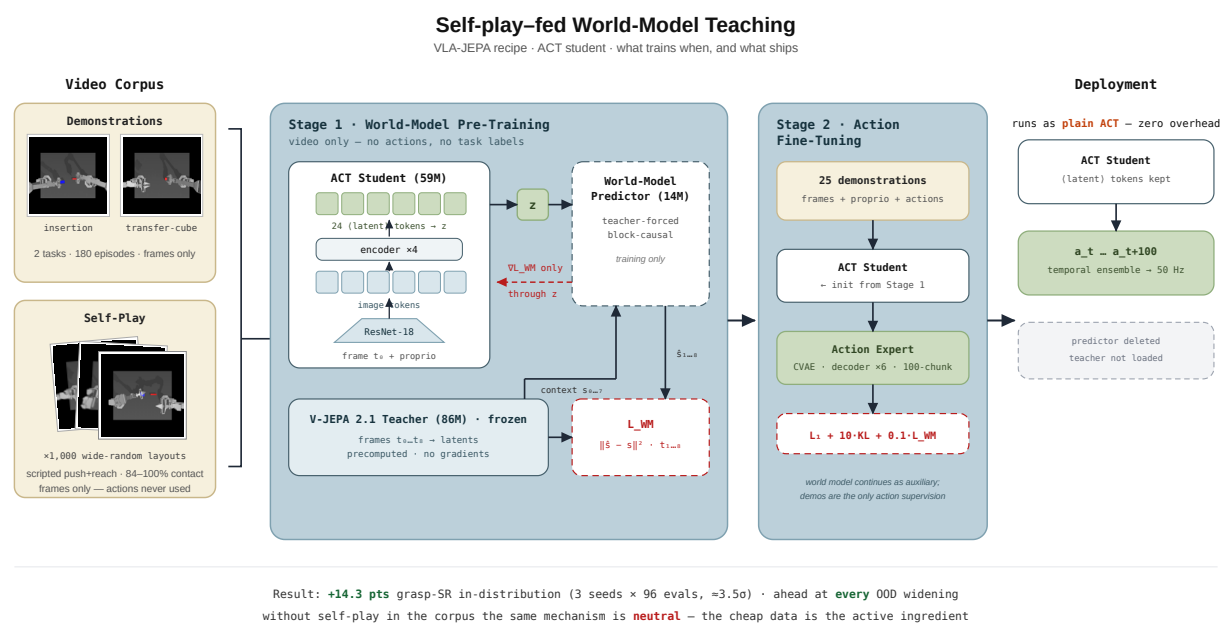
# Embodiment via Self-Play–Fed World-Model Teaching

LeRobot research — aloha sim (bimanual insertion) · VLA-JEPA recipe (arXiv 2602.10098) with an ACT student · June 10–12, 2026 · branch proto/self-play-embodiment

**Question.** Can a world-model module, trained on abundant *cheap* same-embodiment data (self-play), teach a small VLA so it needs fewer demos and generalizes better?

**Headline (3 training seeds × 96 eval seeds).** Feeding **1000 episodes of scripted self-play video** into the world-model corpus makes the K=25-demo policy **+14.3 pts better in-distribution (68.3% vs 54.0% grasp-SR, ≈3.5σ, replicates per-seed)** and ahead at **every** out-of-distribution spawn widening (11/12 cells, sign-test p≈0.003). The same mechanism *without* self-play stays neutral — **the cheap data is the active ingredient**, exactly the original bet.

## The validated recipe



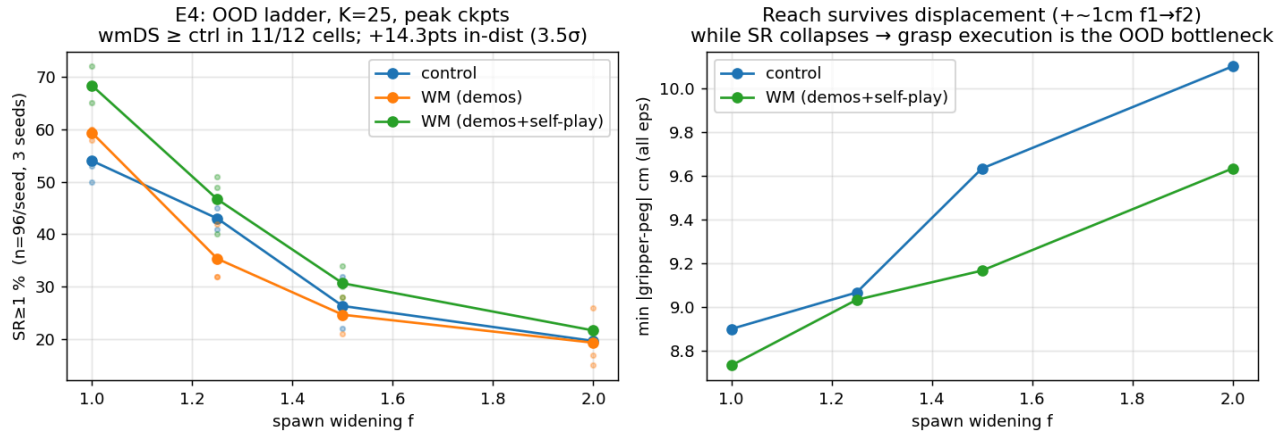
- **Self-play enters as video only** — the WM label is the future itself, so unscripted/imperfect behavior cannot corrupt training (actions never used).
- Faithful VLA-JEPA structure at 59M scale, verified against the paper + the upstream LeRobot port by an **executable self-test gate** (causality probes, leakage tests, mask-equality vs upstream, determinism) — training refuses to start if any check fails.

## Result 1 — the module learns transferable anticipation (representation gate)

| stage-1 corpus    | wide-scene prediction (val/copy ↓) | wide-scene shuffle-z gap ↑ |
|-------------------|------------------------------------|----------------------------|
| demos only        | 2.04 (worse than copying)          | +1.3% (dead)               |
| demos + self-play | 0.574                              | +36.2%                     |

On **held-out wide layouts**, the demos-only module mispredicts and its latent-action channel carries nothing. With self-play in the corpus, anticipation transfers (3.5× better prediction; latent tokens carry scene-specific future info). In-distribution performance unchanged — the diversity costs nothing.

## Result 2 — it converts to policy gains (3 seeds, n=96, peak protocol)



| SR≥1 (grasp), pooled       | in-dist      | 1.25× spawn  | 1.5×         | 2×           |
|----------------------------|--------------|--------------|--------------|--------------|
| control (no WM)            | 54.0%        | 43.0%        | 26.3%        | 19.7%        |
| WM, demos-only corpus      | 59.3%        | 35.3%        | 24.7%        | 19.3%        |
| <b>WM, demos+self-play</b> | <b>68.3%</b> | <b>46.7%</b> | <b>30.7%</b> | <b>21.7%</b> |

- Honest framing: the win is a **higher level everywhere** + small absolute OOD edge — *not* a flatter relative decay (our pre-registered slope prediction failed).
- World model is **dropped at inference** — deployment cost identical to plain ACT.

## Failure analysis — what still breaks (and what doesn't)

- Reach generalizes:** grippers get within ~9 cm of displaced objects even at 2× widening (only ~1 cm worse than in-dist), failures included — measured across all 1,152 OOD episodes after first being spotted by eye in replay datasets.
- Grasp/insert execution is the bottleneck**, in-distribution and OOD alike; secondary mode = late-episode instability past the demo-length tail. ⇒ next gains come from **grasp-diverse demos or grasp-capable play**, not more WM teaching.

## What did NOT work (and saved us from scaling blind)

| attempt                                                                 | verdict                                                                         |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Bottleneck embodiment token on frozen encoder (Fork-2)                  | information-theoretically redundant for a competent policy                      |
| Fine-tuning the vision encoder with dynamics objectives (Fork-1/DynaMo) | injects body, <b>evicts the object</b> ; hurts policy at every demo count       |
| WM teaching with task-redundant corpus (demos only)                     | neutral-to-harmful; dose-dependent interference at small scale                  |
| 2B-scale VLA-JEPA reproduction                                          | no-go: 8×A100 recipe; at our scale the in-dist effect (<1%) is below eval noise |

## Method practices the team can reuse

- **Paper-anchored code + executable self-test gate** — every claim in the paper maps to a runtime assertion; caught 3 silent deviations that had reversed an experimental conclusion.
- **Never compare arms at matched budget** — fixed steps can prove gains but not deficits; use per-arm stopping rules (val-plateau) and peak-over-checkpoints.
- **Closed-loop SR noise:**  $n=24$  swings  $\pm 14$  pts across eval sets;  $n \geq 96 \times \geq 3$  training seeds before believing any gap.
- **Visual failure review** (replayed eval episodes as GUI datasets) re-directed the OOD diagnosis from "reach lock-in" (wrong) to "grasp execution" (measured).

## Recommended next steps

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- **Option A — scale the validated recipe:** bigger student / real-ALOHA corpus arm / more & richer self-play. Evidence basis now exists.
- **Option B — attack the measured bottleneck:** grasp-diverse demonstrations or grasp-capable play (feeds both representation and action channels).
- All code, raw results, and per-seed data: `research/self_play_embodiment/` (FINDINGS.md, scripts/, results/, figures/), commits `53e9cca2d...10800aec0`.

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Generated 2026-06-12 · experiments: aloha sim AlohaInsertion-v0, ACT-style student (59M) + frozen V-JEPA 2.1 teacher · 3 training seeds × 96 paired eval seeds per OOD cell · single RTX 5090